Instructions for ACL-IJCNLP 2021 Proceedings

Anonymous ACL-IJCNLP submission

Abstract

This is the report of the project for the Web Recommender System course held at the University of Copenhagen.

1 Week 6 [Feb 3 - Feb 9]: Familiarize Yourself with the Datasets

1.1 Dataset Description and Import

The dataset used in this project is the "Musical Instruments" dataset from the Amazon Review data 2023 (5-core subset). The dataset consists of 14,000 item ratings from nearly 900 users on approximately 500 items. The data is split into training and test sets, with 80% of the data allocated for training ("train.tsv") and 20% for testing ("test.tsv").

The dataset was imported using the Python library `pandas`(pandas development team, 2020) and its `read_csv` function.

1.2 Data Cleaning and Preprocessing

To ensure the quality of the dataset, we performed the following preprocessing steps:

- Removal of missing values (entries with missing user ID, item ID, or rating).
- Deduplication by keeping the most recent rating when a user rated the same item multiple times.
- Ensuring that all users in the test set are also present in the training set.

In terms of dimensions the effect of the cleaning can be summarized in the Table 1.

Dataset	Before Cleaning	After Cleaning
Train Set	11000	9913
Test Set	3000	1822

Table 1: Effect of the Data Cleaning Process

1.3 Exploratory Data Analysis (EDA) and Statistics

To gain insights into the dataset, we computed the following statistics:

- Distribution of ratings per user in Figure 1
- Distribution of ratings per item in Figure 2
- Overall rating value distribution in Figure 3

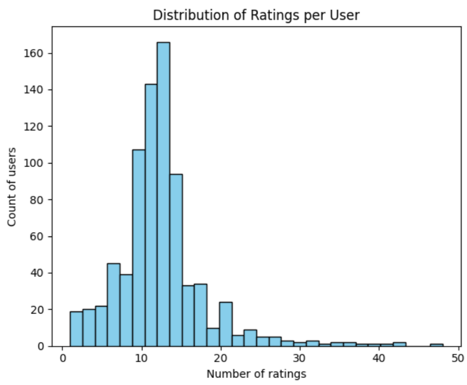

Figure 1: Distribution of Ratings per User

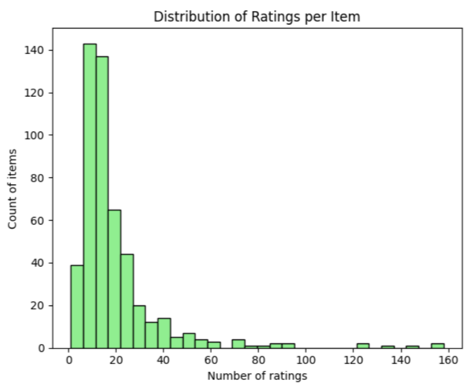

Figure 2: Distribution of Ratings per Item

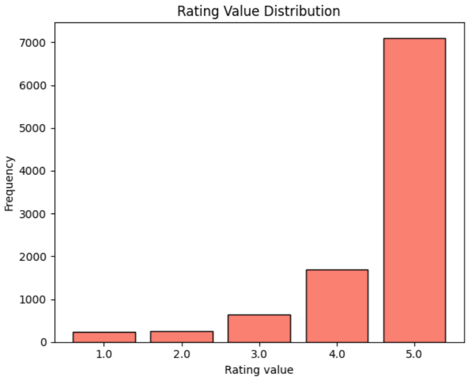

Figure 3: Distribution of Ratings per Vlaue

Also, to show the long-tail distribution of ratings, we plotted the number of ratings per item in Figure 4.

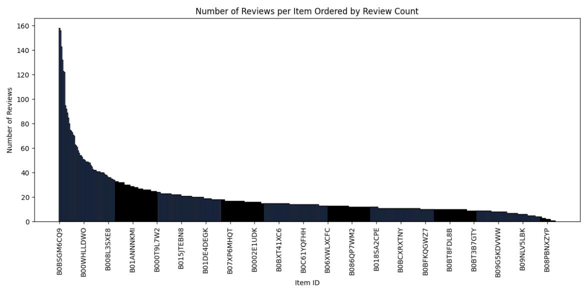

Figure 4: Long tail distribution of rating over all the items

1.4 Highly Rated Items

To identify popular items, we computed the number of ratings ≥ 3 for each item and reported the top 5 most highly rated items in Table 2.

Item ID	High Ratings Count	Average High Rating
B0BSGM6CQ9	153	4.784
B0BPJ4Q6FJ	153	4.804
B09857JRP2	141	4.809
B0BCK6L7S5	120	4.583
B0BTC9YJ2W	119	4.815

Table 2: Top 5 Most Highly Rated Items

1.5 Discussion

The dataset exhibits common characteristics of user-item interactions:

- There is a long-tail distribution where a small number of items receive most of the ratings, while many items have very few ratings.
- The majority of users have rated only a few items, which can lead to sparsity issues in collaborative filtering approaches.
- Certain items dominate the high-rated category, which may bias recommendation systems.

References

The pandas development team. 2020. pandas-dev/pandas: Pandas.